4.1.1. Set Operations

Algorithm

Step 1: Start.
Step 2: Declare two sets A and B
Step 3: Read elements of Set A from the user
Step 4: Read elements of Set B from the user
Step 5: Perform Union operation Union = A ∪ B.
Step 6: Perform Intersection operation Intersection = A ∩ B.
Step 7: Perform Difference operations
Step 8: Difference1 = A − B.
Step 9: Display Set A and Set B
Step 10: Display Union, Intersection, and Difference results
Step 11: Stop

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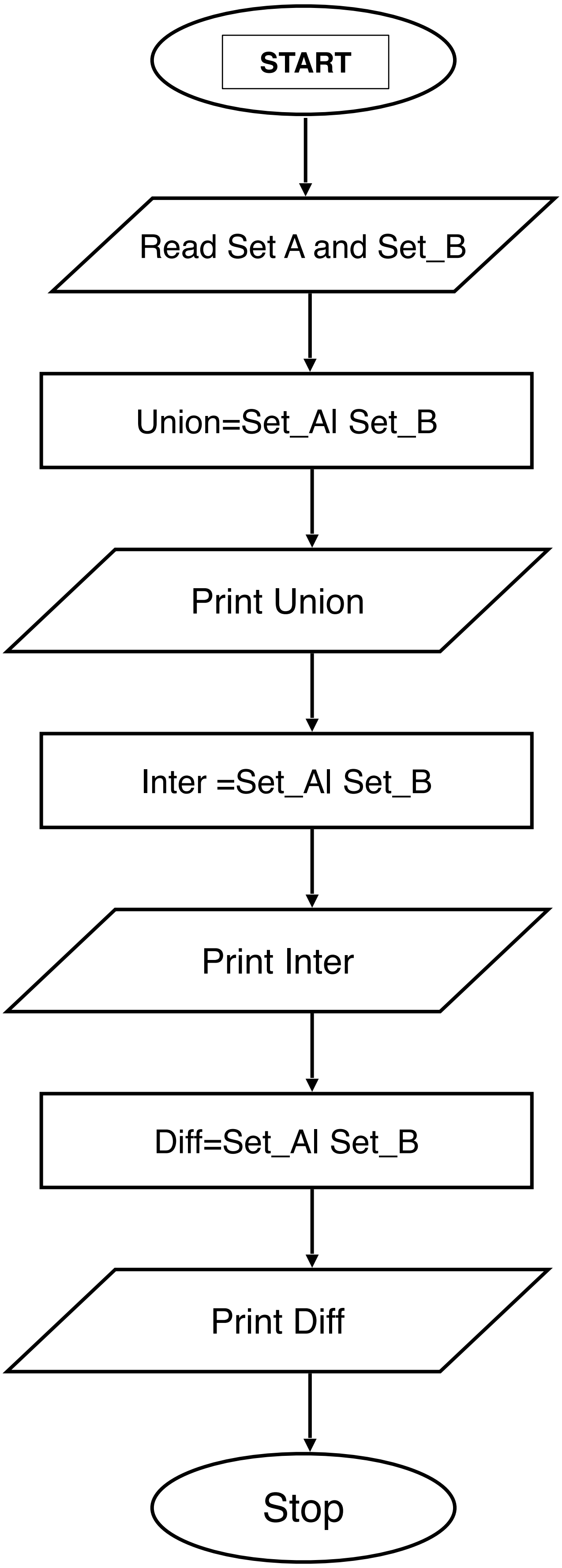
START
Read Set A and Set_B
Union=Set_AI Set_B
Print Union
Inter =Set_AI Set_B
Print Inter
Diff=Set_AI Set_B
Print Diff
Stop

Course

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4.1.1. Set Operations 25:32

Write a Python program to perform union, intersection and difference operations on $Set\ A$ and $Set\ B$.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for $Set\ A$.
- The second input prompts "Set B: " followed by space-separated list of integers for $Set\ B$.

Output Format:

- The first line prints "Union: " followed by the union of $Set\ A$ and $Set\ B$.
- The second line prints "Intersection: " followed by the intersection of $Set\ A$ and $Set\ B$.
- The third line prints "Difference: " followed by the difference of $Set\ A$ and $Set\ B$.

Note:

- If there is no intersection between the two sets, the program prints an empty set, which appears as "set()" in the output.
- Please refer to the visible test cases for better understanding.

Sample Test Cases +

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```python
set_A = set(map(int,input("Set A: ").split()))
set_B=set(map(int,input("Set B: ").split()))
uni_set=set_A | set_B
print("Union:", uni_set)
inter_set=set_A & set_B
print("Intersection:", inter_set)
diff_set=set_A - set_B
print("Difference:", diff_set)
```

Average time	Maximum time
0.004 s	0.006 s
3.50 ms	6.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 6 ms Debug

Expected output	Actual output
Set A: 0 2 4 5 8	Set A: 0 2 4 5 8
Set B: 1 2 3 4 5	Set B: 1 2 3 4 5
Union: {0, 1, 2, 3, 4, 5, 8}	Union: {0, 1, 2, 3, 4, 5, 8}
Intersection: {2, 4, 5}	Intersection: {2, 4, 5}
Difference: {0, 8}	Difference: {0, 8}

Terminal Test cases

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